Output 1:

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex1_bio.py
+-------------------------------------+
| STUDENT BIO CARD                    |
+-------------------------------------+
| Name    : Anu Sharadha
| Age     : 20 years
| Course  : Gen AI
| College : Don Bosco University
| Email   : anushards17@gmail.com
+-------------------------------------+
PS C:\Users\SHARADHA\campuspe>
```

Output 2 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex2_calculator.py
Enter first number: 10
Enter second number:6
Results:
16
4
60
1.6666666666666667
4
1000000
```

Output 3:

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex3a_original.py
Enter a sentence: hello i am sharadha
Original: hello i am sharadha
Characters (with spaces): 19
Characters (without spaces): 16
Total words: 4
UPPERCASE: HELLO I AM SHARADHA
lowercase: hello i am sharadha
Title Case: Hello I Am Sharadha
First word: hello
Last word: sharadha
Reversed: ahdarahs ma i olleh
```

Output 4 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex3a_original.py
Enter a sentence: hello i am sharadha
Original: hello i am sharadha
Characters (with spaces): 19
Characters (without spaces): 16
Total words: 4
UPPERCASE: HELLO I AM SHARADHA
lowercase: hello i am sharadha
Title Case: Hello I Am Sharadha
First word: hello
Last word: sharadha
Reversed: ahdarahs ma i olleh
```

Output 5 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex5_bill_splitter.py
=== BILL SPLITTER ===
Enter total bill amount: 1000
Enter number of people: 4
Enter tax percentage: 15
Enter tip percentage: 10

Total Bill: 1265.0
Amount per person: 316.25
```

Output 6 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex6_grade_calculator.py
=== GRADE CALCULATOR ===
Enter marks of Subject 1: 50
Enter marks of Subject 2: 60
Enter marks of Subject 3: 69
Enter marks of Subject 4: 87
Enter marks of Subject 5: 98

Total Marks: 364
Percentage: 72.8
Grade: B
Result: Pass
```

Output 7:

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex7_temperature_converter.py
Temperature Converter
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Celsius to Kelvin
4. Kelvin to Celsius
5. Fahrenheit to Kelvin
6. Kelvin to Fahrenheit
Fahrenheit: 77.0
```

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex7_temperature_converter.py
Temperature Converter
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Celsius to Kelvin
4. Kelvin to Celsius
5. Fahrenheit to Kelvin
6. Kelvin to Fahrenheit
Enter your choice: 3
Enter Celsius: 30
Kelvin: 303.15
```

Output 8 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex8_leap year cheacker.py"
Enter a year: 2003
2003 is NOT a Leap Year.
Reason: It does not satisfy the leap year conditions.
PS C:\Users\SHARADHA\campuspe> 2004
2004
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex8_leap year cheacker.py"
Enter a year: 2004
2004 is a Leap Year.
Reason: It follows the leap year rule.
```

Output 9:

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex9_movie ticket.py"
Enter your age: 23
Enter day of the week: sunday
Enter number of tickets: 2

----- BILL SUMMARY -----
Base Price: 600
Discount: 120.0
Price After Discount: 480.0
Total Amount: 480.0
```

Output 10 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex10_atm.py

1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Balance: 10000
```

```
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 2
Enter deposit amount: 4000
Deposit successful
New Balance: 14000.0
```

```
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 3
Enter withdraw amount: 6000
Withdrawal successful
New Balance: 8000.0
```

```
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 4

Transaction History:
Deposited 4000.0
Withdrew 6000.0
```

```
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 5
Thank you
```

Output 11 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex11_pattern generator.py"
Choose Pattern:
1. Increasing Numbers
2. Repeating Numbers
3. Reverse Numbers
4. Pyramid Numbers
5. Increasing Stars
6. Decreasing Stars
7. Number Square
Enter choice: 1
Enter height/size: 5
```

```
Enter choice: 1
Enter height/size: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
7. Number Square
Enter choice: 4
Enter height/size: 7
      1
     121
    12321
   1234321
  123454321
 12345654321
1234567654321
```

Output 12 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex12_multiplication Table"
Enter number: 7
Enter range (end): 10
```

```
Multiplication Table of 7
-------------------------
 7 x  1 =   7
 7 x  2 =  14
 7 x  3 =  21
 7 x  4 =  28
 7 x  5 =  35
 7 x  6 =  42
 7 x  7 =  49
 7 x  8 =  56
 7 x  9 =  63
 7 x 10 =  70
-------------------------
```

Output 13 :

```
pData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex13_sum & aver.py"
How many numbers? 5
Enter number 1: 10
Enter number 2: 20
Enter number 3: 30
Enter number 4: 25
Enter number 5: 15
```

```
Results:
Sum: 100
Average: 20.0
Maximum: 30
Minimum: 10
```

Output 14 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex14_Factorial.py
Enter a number: 5
5! = 5 x 4 x 3 x 2 x 1 = 120
PS C:\Users\SHARADHA\campuspe> 0
0
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex14_Factorial.py
Enter a number: 0
0! = 1
PS C:\Users\SHARADHA\campuspe>
```

Output 15 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex15_prime no.py"
Enter a number: 10
10 is NOT a prime number

Enter start range: 5
Enter end range: 20
Prime numbers: 5 7 11 13 17 19
PS C:\Users\SHARADHA\campuspe>
```

Output 16 :

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe c:/Users/SHARADHA/campuspe/ex16_game.py
Guess the number: 10
Too high
Guess the number: 3
Too low
Guess the number: 7
Correct! You guessed the number.
Total attempts (Bonus): 3
PS C:\Users\SHARADHA\campuspe>
```

Output 17 :

```
pData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex17_palindrome .py"
Enter a string: madam
The given string is a Palindrome
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex17_palindrome .py"
Enter a string: civic
The given string is a Palindrome
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex17_palindrome .py"
Enter a string: python
The given string is NOT a Palindrome
PS C:\Users\SHARADHA\campuspe>
```

Output 18:

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex18_calculator fun.py"
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter choice: 1
Enter first number: 12
Enter second number: 8
20
```

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex18_calculator fun.py"
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter choice: 3
Enter first number: 7
Enter second number: 3
21
```

Output 19 :

```
pData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex19_text analy.py"
Enter text: level level

=== TEXT ANALYSIS ===
Words: 2
Vowels: 4
Consonants: 6
Reversed: level level
Palindrome: Yes
Without vowels: lvl lvl
Longest word: level ( 5 letters )
Word Frequency:
level: 2
```

```
Enter text: Python is fun

=== TEXT ANALYSIS ===
Words: 3
Vowels: 3
Consonants: 8
Reversed: nuf si nohtyP
Palindrome: No
Without vowels: Pythn s fn
Longest word: Python ( 6 letters )
Word Frequency:
python: 1
is: 1
fun: 1
```

Output 20 “

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex20_no syst fun.py"
1. Factorial
2. Prime
3. Reverse
4. Sum of digits
Enter your choice: 1
Enter at least 2 numbers (space separated): 12 5
Factorial of 12 = 479001600
Factorial of 5 = 120
```

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex20_no syst fun.py"
1. Factorial
2. Prime
3. Reverse
4. Sum of digits
Enter your choice: 2
Enter at least 2 numbers (space separated): 12 5
12 is Prime: False
5 is Prime: True
```

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex20_no syst fun.py"
1. Factorial
2. Prime
3. Reverse
4. Sum of digits
Enter your choice: 4
Enter at least 2 numbers (space separated): 12 5
Sum of digits of 12 = 3
Sum of digits of 5 = 5
```

```
PS C:\Users\SHARADHA\campuspe> & C:/Users/SHARADHA/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/SHARADHA/campuspe/ex20_no syst fun.py"
1. Factorial
2. Prime
3. Reverse
4. Sum of digits
Enter your choice: 3
Enter at least 2 numbers (space separated): 12 56
Reverse of 12 = 21
Reverse of 56 = 65
```